F9000X_16x7.68TB_nvme_4xNDR400_Nclients1

MLPerf Storage v2.0 Submission

System Name: F9000X_16x7.68TB_nvme_4xNDR400_Nclients1

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Benchmark Version: v2.0

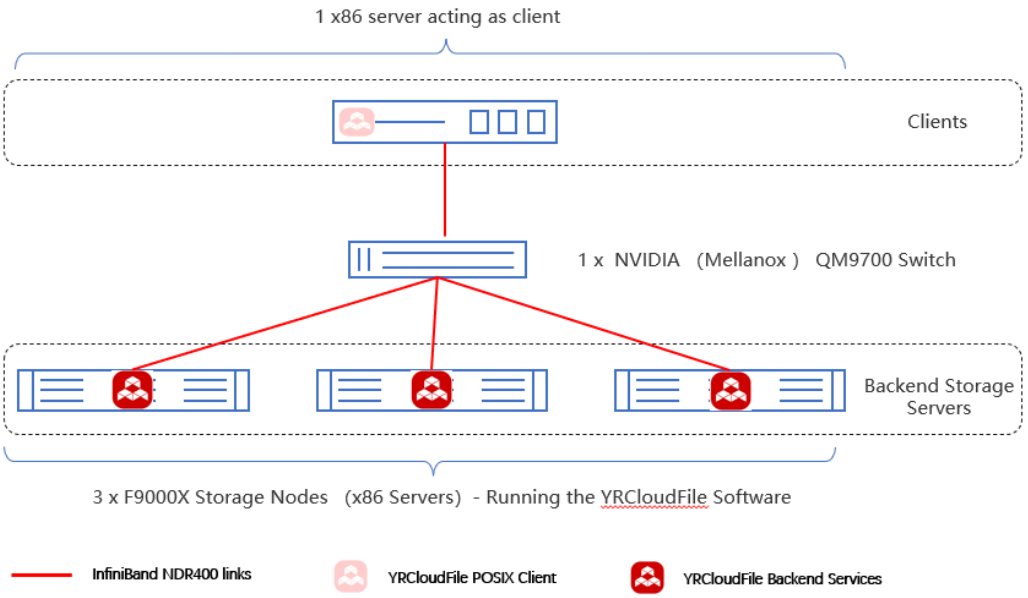
Solution Type: Proprietary - F9000X (parallelle system)

Submission Division: CLOSED

1. Hardware

The hardware is composed of one client and three servers connected to the same switch.

Hardware Diagram



1.1 Client

The client is a commercial x86-architecture server node featuring a dual-socket (2P) Intel-based motherboard, 256 GB of RAM, and default out-of-the-box settings. It communicates with the server over InfiniBand, utilizing two dedicated links exclusively for storage traffic.

The detailed configuration of the client is shown in the following table:

Client Node components	Configuration Description	Client Node Count
Client CPU	• INTEL(R) XEON(R) PLATINUM 8558 x 2	1
Client Memory	• 256GB DDR5-5600 (16x 16GB RDIMMs)	
Client OS Boot Drive	• 250GB SSD x 1	
Client network_type	• Mellanox CX7 Single-port NDR400 InfiniBand HCAs x 2	

1.2 Server

The test environment utilizes a YanRongTech F9000X storage appliance, an x86-based commercial storage solution running the YRCloudFile distributed file system. The system comprises three storage nodes, each configured with: one 480GB SATA SSD for the operating system, sixteen Memblaze 7.68TB NVMe SSDs for data and metadata storage, and four single-port ConnectX-7 400Gbps NDR InfiniBand host channel adapters for high-speed connectivity.

The server communicate with the client through IB:

- 4x single-port NDR400 adapters for a total of 4x NDR400 links

The detailed configuration of the servers is shown in the following table:

Server Node components	Configuration Description	Server Node Count
Server CPU	• Intel(R) Xeon(R) Gold 6530 x 2	3
Server Memory	• 512GB DDR5 4800	
Server OS Boot Drive	• 480GB SATA SSD x 1	
data and metadata drives	• Memblaze 7.68TB nvme SSD x 16	
Server network_type	• Mellanox CX7 Single-port NDR400 InfiniBand HCAs x 4	

1.2.1 Power & Rack Requirements Table

The F9000X Storage Server Node incorporates dual redundant, load-sharing 2000W power supply units (PSUs) in an active/active configuration to ensure continuous operation.Full Power Redundancy: If either PSU fails, the remaining unit instantly assumes the full 2000W power load with zero interruption.Eliminated Single Point of Failure: Both PSUs actively participate in power delivery during normal operation.Hot-Swap Capability: Enables live replacement of PSUs without service interruption.

System Component	Power Supply Unit	Nameplate Rated Power(W)	Design Power(W)	Rack Unit
Storage Node1	Power Supply 1	2000 watts	2000 watts	2

	Power Supply 2	2000 watts		
Storage Node2	Power Supply 1	2000 watts	2000 watts	2
	Power Supply 2	2000 watts		
Storage Node3	Power Supply 1	2000 watts	2000 watts	2
	Power Supply 2	2000 watts		
Totals	–	**12000 watts**	**6000 watts**	6

1.3 Switch

The switch is a commercial NVIDIA （Mellanox） QM9700 The client and server is connected to it.

2. Software

2.1 Client

The configuration of the client-side software is shown in the table below:

Software Type	Software Version
Client OS	Ubuntu 24.04 LTS
Client Kernel	6.8.0-60-generic
yrfs-client	YRCloudFile 6.13.3
Mellanox OFED version	MLNX_OFED_LINUX-24.10-2.1.8.0

2.2 Server

The configuration of the Server-side software is shown in the table below:

Software Type	Software Version
YRCloudFile OS	YRCloudFile 6.13.3
Storage Servers OS	CentOS Linux 7 (Core)
Storage Servers Kernel	5.14.21-20250107.el7.x86_64
Mellanox OFED version	OFED-internal-23.10-4.0.9

3. Settings

3.1 Client

The standard yrfs-client（Version：YRCloudFile 6.13.3）is installed on the client using the default installation

procedure. yrfs-client is used for the client to access the remote YRCloudFile filesystem.

3.2 Server

For detailed configuration information about storage servers, see the table below：

storage_system	Description
vendor_name	YanRongTech
solution_name	YRCloudFile
model_name	F9000X
version	6.13.3
usable_capacity_GiB	157,696

raw_capacity_GiB	315,392

3.3 Switch

No tuning applied.

4. Configuration

4.1 Configure Direct IO

4.1.1 Client Configuration

Edit the configuration file /etc/yrfs/yrfs-client.conf

```
SQL
cluster_addr
=100.10.10.41,100.10.10.45,100.10.10.49,100.10.10.42,100.10.10.46,
100.10.10.50,100.10.10.43,100.10.10.47,100.10.10.51,100.10.10.44,1
00.10.10.48,100.10.10.52
enable_cache = false
enable_multi_channel = true
nr_conn_socks_pernode = 256
rdma_mode=0
mnt_wait_timeout_ms = 16000
```

4.1.2 Storage Backend Configuration

Configure storage through the yrcli command line:

```
SQL
yrcli --config --type=oss --key=enable_read_cache --op=set --
value=false
yrcli --config --type=oss --key=enable_dio_read --op=set --
value=true
```